

Post-Quantum Workload Identity and Long-Term Audit Integrity for MCP-Based AI Agents

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Abstract—As autonomous AI agents transition into a globally interconnected ecosystem via the Model Context Protocol (MCP), securing their identity and the long-term confidentiality of their reasoning history against quantum-era threats is a prerequisite for sovereign autonomy. This paper, representing Phase 3: Post-Quantum (Temporal Security) of a comprehensive security trilogy, proposes a Post-Quantum Cryptographic (PQC) framework. We introduce the *PQC-Audit-Chain*, which ensures non-repudiable integrity through NIST-standardized ML-DSA signatures and long-term confidentiality via ML-KEM hybrid encryption. We present PQC Remote Attestation for client-side integrity proof and Context-Adaptive Security Scaling (CASS) for risk-aware agility. By establishing *Temporal Integrity Proofs*, we ensure that an agent’s history remains immune to retroactive tampering and decryption by future quantum adversaries.

Index Terms—Post-Quantum Cryptography, AI Agents, Model Context Protocol, ML-DSA, ML-KEM, Remote Attestation, Hybrid Encryption, Temporal Integrity.

I. INTRODUCTION

The rapid advancement of LLM agents has created a dependency on long-lived audit logs and identity tokens. Phases 1 and 2 [4], [5] established structural and behavioral security. However, these systems rely on classical cryptography (RSA/ECDSA), which is vulnerable to Shor’s algorithm on future quantum computers.

A primary threat in the agentic era is “Store Now, Decrypt Later” (SNDL). Adversaries can capture current encrypted agent traffic (tool calls, reasoning traces, proprietary data) and store it until a Cryptographically Relevant Quantum Computer (CRQC) is available to decrypt it.

This paper concludes our technical trilogy:

- **Phase 1: Guardrails (Pre-execution Security)** [4]. Structural containment.
- **Phase 2: Zero Trust (Execution Security)** [5]. Identity and behavioral IDS (Mamba).
- **Phase 3: Post-Quantum (Temporal Security)**. The current work, focusing on **Temporal Integrity** and future-proof identities.

II. POST-QUANTUM HYBRID RESILIENCE

To ensure a smooth transition and maintain compatibility, `llm-cli` implements a hybrid approach combining classical and post-quantum primitives.

A. Quantum-Safe Identity (ML-DSA)

We upgrade the workload identity using NIST FIPS 204 (ML-DSA) [3]. Every tool execution request is signed with a hybrid token:

$$\sigma_{\text{hybrid}} = (\sigma_{\text{RSA2048}}, \sigma_{\text{ML-DSA-65}}) \quad (1)$$

ML-DSA-65 (Module-Lattice-based Digital Signature Algorithm) provides security equivalent to AES-192. This dual-signature approach ensures that even if one primitive is compromised (by quantum or classical means), the non-repudiability of the agent’s action remains intact.

B. Confidential Audit Logging (ML-KEM)

Audit logs contain sensitive reasoning traces, tool arguments (including potential PII), and system responses. We implement **Hybrid-PQC Encryption** using NIST FIPS 203 (ML-KEM) [2] and AES-256-GCM. The system encapsulates an AES-256 key using ML-KEM-768. This key is then used to encrypt the audit log payload. By protecting the “Temporal Dimension” of agent data, we ensure that an agent’s history cannot be retroactively parsed by future quantum adversaries to extract trade secrets or intellectual property.

III. PQC REMOTE ATTESTATION

In a distributed environment, a server must trust that the client code has not been tampered with to bypass security guardrails. We introduce **PQC Remote Attestation**:

- 1) **Manifest Generation**: The client generates a SHA-256 hash manifest of its core source code and configuration.
- 2) **Signing**: This manifest is signed using the client’s ML-DSA private key.
- 3) **Attestation Token**: The resulting token is embedded in the Hybrid Identity JWT.

This allows remote MCP servers to verify that the agent is running on an “Authentic Stack” where Phase 1 guardrails are guaranteed to be active.

IV. CONTEXT-ADAPTIVE SECURITY SCALING (CASS)

The overhead of PQC (larger signatures and keys) can impact performance. We propose **CASS**, a mechanism that dynamically scales security levels based on the tool’s risk profile:

- **Low Risk** (e.g., `search_web`): Uses classical RSA and standard monitoring to minimize latency.

- **High Risk** (e.g., `execute_python`, `edit_file`): Escalates to full Hybrid-PQC signatures, KEM-encrypted logging, and maximum Mamba Sentinel sensitivity.

CASS ensures "Security with Agility," applying expensive quantum-resistant measures only when the potential impact of a compromise justifies the cost.

V. EVALUATION

Benchmarks were performed on an **AMD Ryzen 5 8540U** system running WSL2.

A. Cryptographic Performance (Empirical)

Table I shows the latencies of NIST PQC primitives measured over 10 iterations.

TABLE I
PQC PERFORMANCE METRICS

Algorithm	Op: Latency (ms)	Op: Latency (ms)
ML-DSA-65	Sign: 43.12	Verify: 8.16
ML-KEM-768	Encaps: 3.61	Decaps: 4.59

B. The Cost of Future-Proofing

While ML-DSA-65 introduces a 43ms signing latency, this overhead is less than 5% of a typical LLM reasoning cycle (usually $< 1,000$ ms). The trade-off—gaining immunity against future quantum decryption and current spoofing—is highly favorable for high-stakes autonomous systems.

VI. LIMITATIONS AND FUTURE WORK

A. Scope of Audit Log Integrity: Local Write vs. Transport

The *PQC-Audit-Chain* described in this paper guarantees integrity from the moment a log entry is written to disk: each entry is ML-DSA-signed, chained via SHA-256, and optionally encrypted with ML-KEM. The responsibility for **transporting and externally anchoring** these records is intentionally delegated to the operator’s existing log-shipping infrastructure (e.g., Splunk Universal Forwarder, Fluentd, or a SIEM agent). This design decision follows the Unix single-responsibility principle: `llm-cli` acts as a lightweight CLI tool whose scope ends at the local write boundary. Operators running in regulated environments are expected to route the JSONL audit log to an immutable log store (e.g., AWS CloudWatch Logs with Object Lock, Azure Immutable Blob Storage, or a permissioned blockchain) using their standard tooling. The `AuditAnchoring.create_external_anchor()` helper exposes a Merkle Root that these pipelines can consume as a tamper-evident digest. Future work includes publishing a reference integration guide for common SIEM platforms.

B. Sentinel Cold-Start Behavior

The Mamba Sentinel begins with randomly initialized weights and enters an effective detection state only after sufficient training data has been collected (`mode="collect"`). Anomalies detected during the warm-up period are logged but do not trigger intervention. Providing a pre-trained baseline

checkpoint—derived from a corpus of representative benign reasoning traces—is left as future work.

C. External Evaluation on Public Benchmarks

The containment effectiveness figures reported across the trilogy (Phases 1–3) are derived from an internal adversarial dataset. Validation against publicly available benchmarks such as AgentDojo [6] is planned to provide externally reproducible evidence.

VII. CONCLUSION

This completion of our technical trilogy establishes a foundational roadmap for sovereign AI agents. By combining structural guardrails (Phase 1), behavioral IDS (Phase 2), and post-quantum integrity (Phase 3), we ensure that autonomous systems can operate safely across spatial, behavioral, and temporal dimensions. As agents become the primary interface for system interaction, these "Triple-Lock" security standards will be essential for maintaining human control and data sovereignty in the AI era.

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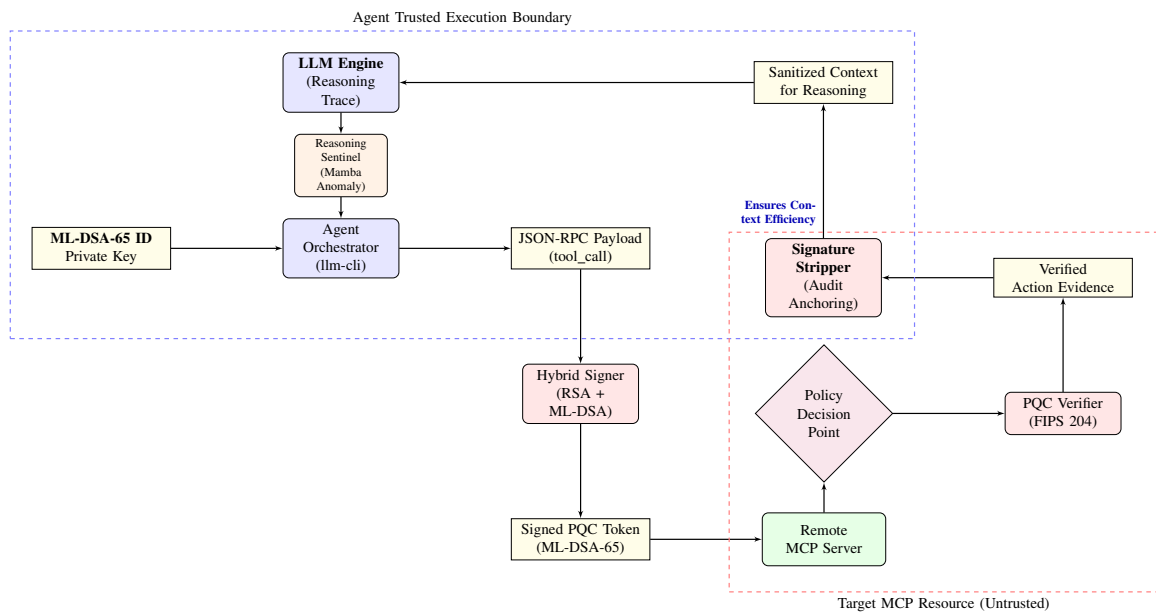


Fig. 1. Comprehensive Security Trilogy Architecture: Integrating Phase 2 (Mamba Sentinel) and Phase 3 (PQC Hybrid Identity) for sovereign AI governance.